

# Class Agnostic Time Series with tsbox :: CHEAT SHEET



## Basics

### IDEA

tsbox provides a time series toolkit which:

1. works identically with most time series **classes**
2. handles regular and irregular **frequencies**
3. **converts** between classes and frequencies

Most functions in tsbox have the same structure:

function starts with **ts\_**

first argument is any **ts-boxable object**

```
a <- ts_pc(AirPassengers)
```

returns a ts-boxable object of the same class as input

### COMBINE TIME SERIES

collect time series of **all classes** and **frequencies** as multiple time series



```
ts_c(mdeaths, austres)
```

combine time series to a new, single time series (first series wins if overlapping)



```
ts_bind(mdeaths, austres)
```

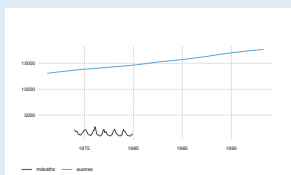
like `ts_bind`, but extra- and retropolate, using growth rates



```
ts_chain(mdeaths, austres)
```

### PLOT AND SUMMARIZE

Plot time series of **all classes** and **frequencies**



```
ts_plot(mdeaths, austres)
ts_ggplot(mdeaths, austres)
```

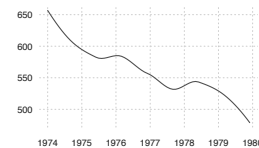
```
ts_summary(ts_c(mdeaths, austres))
```

|   | id      | obs | diff | freq  | start         | end        |
|---|---------|-----|------|-------|---------------|------------|
| 1 | mdeaths | 72  | 1    | month | 12 1974-01-01 | 1979-12-01 |
| 2 | austres | 89  | 3    | month | 4 1971-04-01  | 1993-04-01 |

## Helper Functions

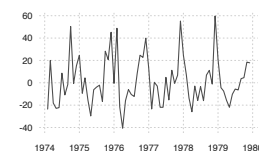
Transform time series of **all classes** and **frequencies**

### TRANSFORM



**ts\_trend()**: Trend estimation based on loess

```
ts_trend(fdeaths)
```



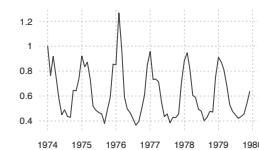
**ts\_pc()**, **ts\_pcy()**, **ts\_pca()**, **ts\_diff()**, **ts\_diffy()**: (annualized) Percentage change rates or differences to previous period, year

```
ts_pc(fdeaths)
```



**ts\_scale()**: normalize mean and variance

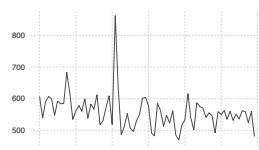
```
ts_scale(fdeaths)
```



**ts\_index()**: Index, based on levels

**ts\_compound()**: Index, based on growth rates

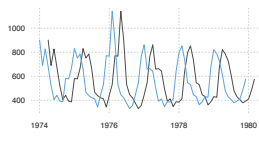
```
ts_index(fdeaths, base = 1976)
```



**ts\_seas()**: seasonal adjustment using X-13

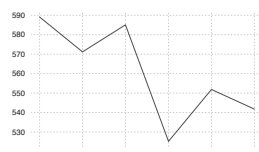
```
ts_seas(fdeaths)
```

### SPAN AND FREQUENCY



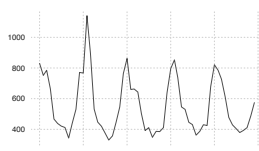
**ts\_lag()**: Lag or lead of time series

```
ts_lag(fdeaths, 4)
```



**ts\_frequency()**: convert to frequency

```
ts_frequency(fdeaths, "year")
```



**ts\_span()**: filter time series for a time span.

```
ts_span(fdeaths, "1976-01-01")
ts_span(fdeaths, "-5 year")
```

## Class Conversion

tsbox is built around a set of converters, which convert time series of the following **supported classes** to each other:

| converter function                                  | ts-boxable class       |
|-----------------------------------------------------|------------------------|
| <code>ts_ts()</code>                                | ts, mts                |
| <code>ts_data.frame()</code> , <code>ts_df()</code> | data.frame             |
| <code>ts_data.table()</code> , <code>ts_dt()</code> | data.table             |
| <code>ts_tbl()</code>                               | df_tbl, "tibble"       |
| <code>ts_xts()</code>                               | xts                    |
| <code>ts_zoo()</code>                               | zoo                    |
| <code>ts_tibbltime()</code>                         | tibbltime              |
| <code>ts_timeSeries()</code>                        | timeSeries             |
| <code>ts_tsibble()</code>                           | tsibble                |
| <code>ts_tslist()</code>                            | a list with ts objects |

## Time Series in data frames

### LONG STRUCTURE

Default structure to store multiple time series in long data frames (or data tables, or tibbles)

```
ts_df(ts_c(fdeaths, mdeaths))
```

| id      | time       | value |
|---------|------------|-------|
| fdeaths | 1974-01-01 | 901   |
| fdeaths | 1974-02-01 | 689   |
| fdeaths | 1974-03-01 | 827   |
| ...     | ...        | ...   |

### AUTO-DETECT COLUMN NAMES

tsbox auto-detects a *value*-, a *time*- and zero, one or several *id*-columns. Alternatively, the *time*- and the *value*-column can be explicitly named **time** and **value**.

**ts\_default()**: standardize column names in data frames

### RESHAPE

**ts\_wide()**: convert default long structure to wide

**ts\_long()**: convert wide structure to default long

### USE WITH PIPE

tsbox plays well with tibbles and with `%>%`, so it can be easily integrated into a dplyr/pipe workflow

```
library(dplyr)
ts_c(fdeaths, mdeaths) %>%
  ts_tbl() %>%
  ts_trend() %>%
  ts_pc()
```

pass return value as first argument to the next function